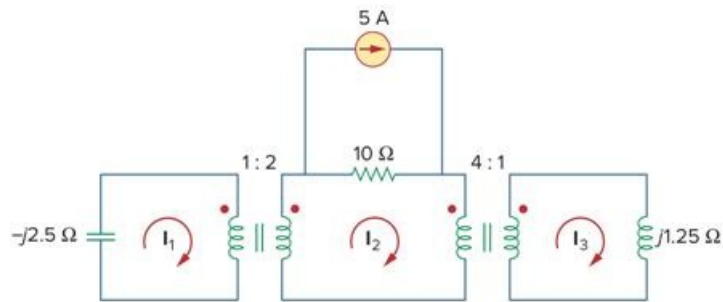


Problem

Find the mesh currents in the circuit of Fig. 13.128

Figure 13.128



Step-by-step solution

Step 1 of 8

Refer to Figure 13.128 from the textbook.

Consider the number of turns ratio of the two transformers as:

$$n_1 = \frac{1}{2}$$

$$n_2 = \frac{4}{1}$$

Step 2 of 8

Calculate the reflected impedance of $j1.25 \Omega$ on the middle side, Z'_3 .

$$Z'_3 = Z_3 (n_2)^2$$

Substitute $j1.25 \Omega$ for Z_3 and 4 for n_2 .

$$\begin{aligned} Z'_3 &= j1.25(4)^2 \\ &= 16(j1.25) \\ &= j20 \Omega \end{aligned}$$

Step 3 of 8

Calculate the reflected impedance of $-j2.5 \Omega$ on the middle side, Z'_2 .

$$Z'_2 = \frac{Z_1}{(n_1)^2}$$

Substitute $-j2.5 \Omega$ for Z_1 and $\frac{1}{2}$ for n_1 .

$$\begin{aligned} Z'_2 &= \frac{-j2.5}{\left(\frac{1}{2}\right)^2} \\ &= (-j2.5)(4) \\ &= -j10 \Omega \end{aligned}$$

Sketch the modified circuit as shown in Figure 1.

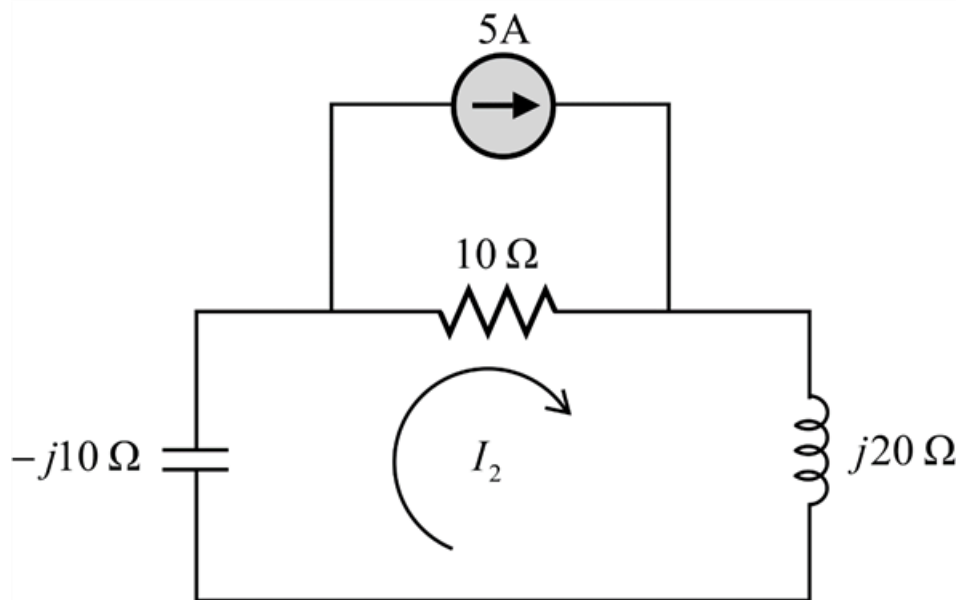


Figure 1

Since $10\ \Omega$ is in parallel with the current source, replace the current source by equivalent voltage source with $10\ \Omega$ in series with it as shown in Figure 2.

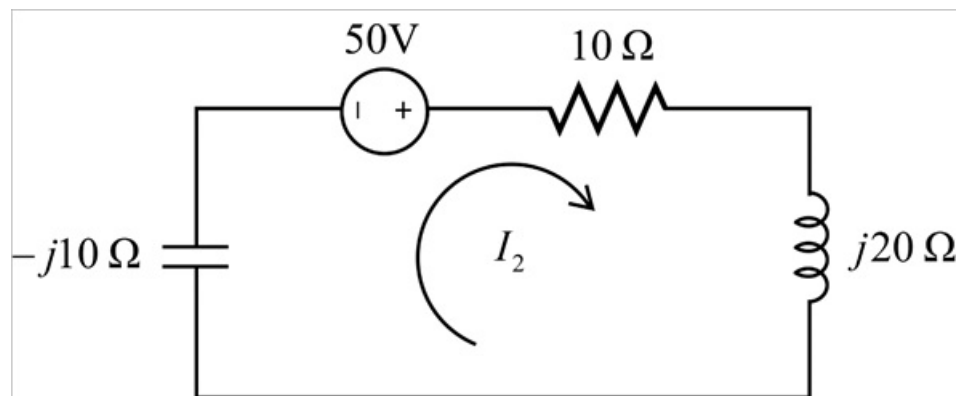


Figure 2

[Comments \(1\)](#)



Anonymous

Much easier to use the current divider equation. $I_2 = 5 * (10 / (10 + j10))$

Apply Kirchhoff's Voltage Law in the the circuit shown in Figure 2.

$$0 = -50 + 10I_2 + j20I_2 - j10I_2$$

$$0 = -50 + (10 + j10)$$

$$I_2 = \frac{50}{10 + j10}$$

$$I_2 = 2.5 - j2.5$$

$$I_2 = 3.535 \angle -45^\circ \text{ A}$$

Step 7 of 8

For the first transformer, the ratio of the mesh currents is,

$$\frac{I_1}{I_2} = \frac{1}{n_1}$$

$$I_1 = \frac{1}{n_1} I_2$$

Substitute $3.535 \angle -45^\circ \text{ A}$ for I_2 and $\frac{1}{2}$ for n_1 .

$$\begin{aligned} I_1 &= \frac{1}{\left(\frac{1}{2}\right)} (3.535 \angle -45^\circ) \\ &= 7.071 \angle -45^\circ \text{ A} \end{aligned}$$

Step 8 of 8

For the second transformer, the ratio of the mesh currents is,

$$\frac{I_2}{I_3} = \frac{1}{n_2}$$

$$I_3 = n_2 I_2$$

Substitute $3.535 \angle -45^\circ \text{ A}$ for I_2 and 4 for n_2 .

$$\begin{aligned} I_3 &= (4)(3.535 \angle -45^\circ) \\ &= 14.142 \angle -45^\circ \text{ A} \end{aligned}$$

Therefore, the mesh currents, I_1 is $7.071 \angle -45^\circ \text{ A}$, I_2 is $3.535 \angle -45^\circ \text{ A}$ and I_3 is $14.142 \angle -45^\circ \text{ A}$.